

Shreepriya Karane

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Professional Summary

AI Engineer and Computer Science graduate with experience building scalable AI-powered systems, distributed backend architectures, and machine learning applications using **Python, FastAPI, Java, LangChain, SQL, and AWS**. Strong foundation in **machine learning, NLP, distributed systems, data structures, algorithms, and analytical problem solving** with demonstrated ability to develop production-grade AI applications, optimize large-scale processing pipelines, and conduct experimentation-driven engineering. Passionate about **research, intelligent systems, and state-of-the-art AI technologies** with experience collaborating in Agile teams to deliver scalable and reliable software solutions.

Technical Skills

Programming Languages: Python, Java, JavaScript (ES6), TypeScript, SQL

Core CS Concepts: Data Structures & Algorithms, Object-Oriented Design, Distributed Systems, Complexity Analysis, System Design, Problem Solving

AI / Machine Learning: Machine Learning, NLP, Generative AI, LLM Applications, LangChain, Prompt Engineering, Scikit-learn, Vector Databases, Feature Engineering, Model Evaluation

Research & Analytics: Experimental Analysis, Data Processing, Predictive Modeling, Analytical Problem Solving, AI System Optimization

Backend Engineering: FastAPI, Spring Boot, REST APIs, Microservices, Distributed Processing Systems

Cloud & DevOps: AWS (EC2, S3), Docker, Kubernetes, Jenkins, CI/CD

Databases: MySQL, MongoDB, SQL Optimization, Relational Databases

Frontend: React.js, HTML, CSS

Tools & Practices: Git, Postman, PyTest, Unit Testing, Debugging, Agile Methodologies, Documentation, Code Reviews

Professional Experience

AI Developer Intern

Sept 2025 – Mar 2026

Smartkeen Tech, Bengaluru

- Engineered scalable AI-powered backend systems and distributed REST APIs using **Python, FastAPI, SQL, and cloud-native workflows** supporting enterprise-scale production applications.
- Improved backend **throughput by 25–30%** and **reduced API latency by 20–25%** through asynchronous execution, query optimization, caching strategies, and efficient request orchestration.
- Built modular microservices and optimized distributed data-processing pipelines improving scalability, fault tolerance, and data consistency by **30–40%**.
- Conducted experimentation-driven optimization and system evaluation workflows to improve inference efficiency, reliability, and production stability across AI services.
- Collaborated with engineering teams in Agile development environments to debug production systems, improve maintainability, and deliver scalable AI-driven software solutions.
- Designed and optimized scalable AI data-processing workflows handling large-scale structured and unstructured datasets, improving processing efficiency, system reliability, and backend scalability for production AI applications.

Independent AI Research & Engineering Project

Apr 2025 – Jul 2025

- Architected a scalable Generative AI platform supporting semantic retrieval, intelligent response generation, and distributed document-processing workflows using modern RAG architectures.
- Developed modular backend systems using **Python, FastAPI, LangChain, and vector databases** enabling efficient retrieval pipelines and scalable AI inference workflows.
- Achieved **sub-2s inference latency** through asynchronous processing, vector indexing optimization, caching mechanisms, and backend performance tuning.
- Applied NLP concepts, prompt engineering, experimentation workflows, and AI system evaluation techniques to improve response quality, retrieval accuracy, and scalability.
- Designed distributed query-processing workflows for large-scale document ingestion and semantic search improving responsiveness, modularity, and production reliability.

- Developed production-grade REST APIs achieving **sub-500ms response times** under concurrent workloads using scalable backend engineering practices.
- Optimized relational database performance by **35%** through SQL query optimization, indexing strategies, and efficient data-access workflows.
- Built scalable frontend and backend modules using **React.js, TypeScript, JavaScript, REST APIs, and MySQL** improving maintainability, responsiveness, and application reliability.

Projects

DocMind – Generative AI Document Assistant

Python, FastAPI, LangChain, FAISS, LLM APIs

GitHub ↗

- Built a scalable AI-powered document assistant supporting semantic retrieval, vector search, intelligent response generation, and distributed RAG-based processing workflows.
- Developed modular FastAPI backend services and AI orchestration pipelines improving scalability, maintainability, and retrieval efficiency.
- Achieved **sub-2s response latency** through vector indexing optimization, asynchronous execution, and caching-based performance improvements.

Stress Detection ML System

Python, Scikit-learn, Machine Learning

- Built a machine learning classification system achieving **85% accuracy** using optimized preprocessing, feature engineering, and model-training workflows.
- Conducted model evaluation, validation, and feature-selection experiments improving predictive reliability and classification performance.
- Applied analytical problem solving and machine learning methodologies to address real-world predictive analytics challenges.

Incentive Management Application – Client Project

Java, Spring Boot, React.js, TypeScript, REST APIs, MySQL

GitHub ↗

- Developed scalable backend services and modular REST APIs using **Java, Spring Boot, React.js, and MySQL** for a production-oriented multi-tier client application.
- Optimized database workflows, API integration, and modular business logic improving backend maintainability, query efficiency, and application reliability across enterprise workflows.

Automotive Data Analysis Dashboard

Python, Pandas, Streamlit, Data Analytics

GitHub ↗

- Developed a scalable analytics dashboard supporting automated data ingestion, preprocessing, transformation, and visualization workflows.
- Built modular data-processing pipelines improving reporting efficiency, analytical consistency, and maintainability for structured datasets.
- Generated actionable analytical insights through exploratory data analysis and visualization techniques supporting data-driven decision making.

Education

B.E. in Computer Science and Engineering

Bapuji Institute of Engineering and Technology

2021 – 2025

Achievements

- Winner – Byte Talks Coding Competition
- Winner – Planathon Debate Competition